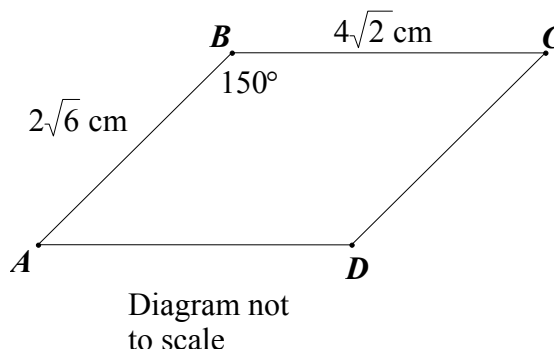


Year 12 Trial HSC Examination - Mathematics (2U) 2006

Question 1

	Marks
(a) Find the value of $\log_6 12$ correct to two decimal places.	2
(b) The investment value of a stamp collection was originally \$24 800. If the value of the collection compounds at a rate of 5.2% annually, find its estimated value at the end of 10 years. Give your answer correct to the nearest \$100.	2
(c) The volume V of a cylinder with a base radius r and height h is given by the formula $V = \pi r^2 h$. Find the height of a cylinder when the volume is 480 cm^3 and base radius 9.5 cm. Give your answer correct to the nearest millimetre.	2
(d) Solve the equation $3(2x + 1) - 2(3 - x) = 53$.	3
(e) Find the exact length of the longer diagonal in parallelogram $ABCD$.	3



Question 2 (Start a new page)

	Marks
The coordinates of the points E , F and G are $(0,3)$, $(6,0)$ and $(7,-4)$ respectively.	
(a) Draw a neat sketch, clearly showing the above information and show that the line k which is parallel to EF and passes through point G , has the equation $x + 2y + 1 = 0$.	3
(b) Find the coordinates of the point H where line k meets the x -axis. Clearly mark point H and the position line k on your diagram.	2
(c) Find the exact length of EF .	2
(d) Find perpendicular distance from point F to line GH .	2
(e) Find the exact area of quadrilateral $EFGH$.	3

Question 3 (Start a new page)

	Marks
(a) Differentiate with respect to x :	
(i) $\frac{4x^3\sqrt{x+5}}{x^2}$,	2
(ii) $(8-3x)^5$.	2
(b) Find the equation of the tangent to $y = xe^{-x}$ at the point where $x = 2$. Write your answer in general form.	4
(c) Differentiate $y = \frac{\cos x}{1 + \sin x}$,	4
and hence show that $\frac{dy}{dx} = \frac{-1}{1 + \sin x}$.	

Question 4 (Start a new page)

	Marks
(a) Evaluate	
(i) $\int_0^4 \frac{dx}{3x+2}$,	2
(ii) $\int_{-1}^2 \cos\left(\frac{\pi}{2}x\right) dx$.	2
(b) (i) Find the coordinates of the stationary points on the curve $y = (x+1)(x^2-8)$ and determine their nature.	5
(ii) Draw a neat half page sketch of $y = (x+1)(x^2-8)$ in the domain $-4 \leq x \leq 4$. On your diagram clearly indicate the coordinates and the positions of the intercepts with the coordinate axes and the stationary points.	3

Question 5 (Start a new page)

	Marks
(a) Town A is 4 km from town P and its bearing from town P is $030^\circ T$. Town B is due south of town A and 6 km from town P .	
(i) Draw a neat sketch that clearly illustrates the above information.	2
(ii) Find the distance between towns A and B . Give your answer correct to the nearest kilometre.	4
(b) (i) Find the intersection points of the line $y = 2x$ and the parabola $y = 6x - x^2$.	2
(ii) Find the area bounded by the line $y = 2x$ and the parabola $y = 6x - x^2$.	4

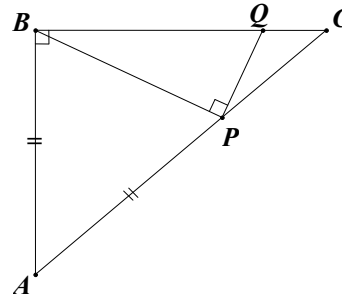
Question 6 (Start a new page)

	Marks
(a) A particle is moving along a straight line. The particle is initially at the origin O and at time t minutes its velocity, v m/min, is given by $v = 1 + 2 \sin\left(\frac{\pi}{4}t\right)$.	
(i) Sketch the graph of v against t for $0 \leq t \leq 8$.	2
(ii) Find the position of the particle when it first reaches its maximum speed.	4
(b) The gradient function of a curve is given by $\frac{dy}{dx} = 3 + \frac{10}{\sqrt{x}}$. Find the equation of the curve if it passes through the point $P(4,9)$.	3
(c) (i) Write down the discriminant (Δ) of the quadratic equation $x^2 + mx + m = 0$.	1
(ii) Hence, or otherwise, find the values of k for which the line $y = x + k$ and the curve $y = \frac{x}{x+1}$ do not intersect.	2

Question 7 (Start a new page)

Marks

- (a) $\triangle ABC$ and $\triangle BPQ$ are right-angled triangles and $AB = AP$.

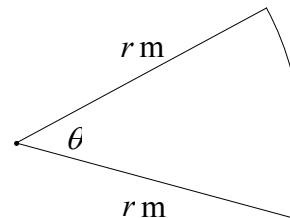


- (i) Copy the diagram onto your answer sheet and prove that $\angle CBP = \angle CPQ$.
(Hint: Let $\angle CBP = \alpha^\circ$) 3
- (ii) Hence prove that $PC^2 = BC \times QC$ 3
- (b) A mass of wire cable is wound around a winch. The mass, M kg, of wire on the winch, at time t minutes, is given by the formula $M = 400 - 25\sqrt{t + 100}$.
- (i) Find the initial mass of wire on the winch. 2
- (ii) Find the time needed to remove all the wire from the winch. 2
- (iii) Find the initial rate at which the wire is being removed from the winch. 2

Question 8 (Start a new page)

Marks

- (a) John starts work on an annual salary of \$38 500. On the last day of each year for the first 10 years of employment, he will receive an annual salary increase of \$1 200. For his remaining years of employment he will receive an annual increase of \$1 650.
- (i) What will John's annual salary be during his 6th year of employment? 2
- (ii) What will John's annual salary be during his 15th year of employment? 2
- (iii) How much will John's total earnings amount to at the end of his 25th year of employment? 2
- (b) A wire framework enclosing an area of 144 m^2 is to be made in the shape of a sector of a circle (see diagram). The length of wire required is L metres where the circle has radius r metres and the sector angle is θ radians.



- (i) Show that $L = 2r + \frac{288}{r}$. 2
- (ii) Find the minimum length of wire needed to make the framework. 4

Question 9 (Start a new page)

	Marks
(a) Point T is 6 cm from the centre O of a circle with radius 3 cm. Tangents drawn from point T touch the circle at points M and N and the tangents are perpendicular to the radii at these points of contact.	
(i) Find the exact size of $\angle MOT$.	2
(ii) Find the exact area of the minor sector MON .	3
(b) (i) Sketch the curve $y = 2 + \sqrt{x}$ and clearly shade the area bounded by the curve, the coordinate axes and the line $x = 9$.	2
(ii) Find the exact volume of the solid formed when the above area is rotated one revolution about the y-axis .	5

Question 10 (Start a new page)

	Marks
(a) In order to enter a university each student is required to sit for a theory examination. If a student passes the theory examination they are then required to sit for a practical examination. It is estimated that the probability for a student to pass the theory examination is 0.8 and to pass the practical examination is 0.75.	
(i) Find the probability that a student chosen at random will gain entry to the university.	2
(ii) Find the probability that if two students chosen at random only one will gain entry to the university.	2
(b) At 6am a population was observed to be undergoing exponential growth and after time t hours the number of individuals N_G was given by $N_G = 100e^{0.3t}$. When the population reached 500, a virus attacked the population and from that time the population underwent exponential decay so that t hours after the entry of the virus, the number of individuals N_D present was given by $N_D = 500e^{kt}$ for some constant $k < 0$. When the population fell to its original value it was decreasing at a rate of 15 individuals per hour.	
(i) At what time of the day did the population reach 500? Give your answer correct to the nearest minute.	3
(ii) Find the value of the constant k .	2
(iii) At what time of the day did the population return to its original value?	3

This is the END of the examination paper

Year 12 Trial HSC Examination - Mathematics (2U) 2006 - Solutions

Question 1

- (a) Find the value of $\log_6 12$ correct to two decimal places.

$$\begin{aligned}\log_6 12 &= \frac{\log 12}{\log 6} \\ &= 1.39 \text{ (to 2 d.p.)}\end{aligned}$$

- (b) The investment value of a stamp collection was originally \$24 800. If the value of the collection compounds at a rate of 5.2% annually, find its estimated value at the end of 10 years. Give your answer correct to the nearest \$100.

$$\begin{aligned}\text{Value} &= \$24800 \left(1 + \frac{5.2}{100}\right)^{10} \\ &= \$41200 \text{ (to the nearest \$100)}\end{aligned}$$

- (c) The volume V of a cylinder with a base radius r and height h is given by the formula $V = \pi r^2 h$. Find the height of a cylinder when the volume is 480 cm^3 and base radius is 9.5 cm . Give your answer correct to the nearest millimetre.

$$\begin{aligned}V &= \pi r^2 h \\ h &= \frac{V}{\pi r^2} \\ &= \frac{480}{\pi \times 9.5^2} \\ \text{height} &= 1.7 \text{ cm (to nearest mm)}\end{aligned}$$

- (d) Solve the equation $3(2x + 1) - 2(3 - x) = 53$.

$$\begin{aligned}3(2x + 1) - 2(3 - x) &= 53 \\ 6x + 3 - 6 + 2x &= 53 \\ 8x - 3 &= 53 \\ 8x &= 56 \\ x &= 7\end{aligned}$$

- (e) Find the exact length of the longer diagonal in parallelogram $ABCD$.

$$\begin{aligned}AC^2 &= (2\sqrt{6})^2 + (4\sqrt{2})^2 - 2(2\sqrt{6})(4\sqrt{2})\cos 150^\circ \\ &= 24 + 32 - 16\sqrt{12} \times \left(-\frac{\sqrt{3}}{2}\right) \\ &= 56 + 8\sqrt{36} \\ &= 104 \\ \text{length } AC &= \sqrt{104} \text{ cm } (= 2\sqrt{26} \text{ cm})\end{aligned}$$

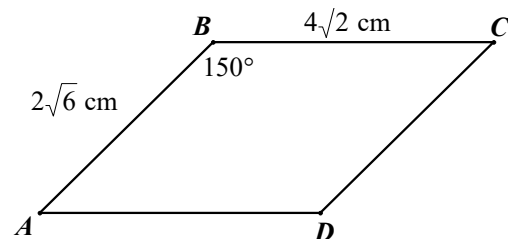


Diagram not to scale

Marks
2

2

2

3

3

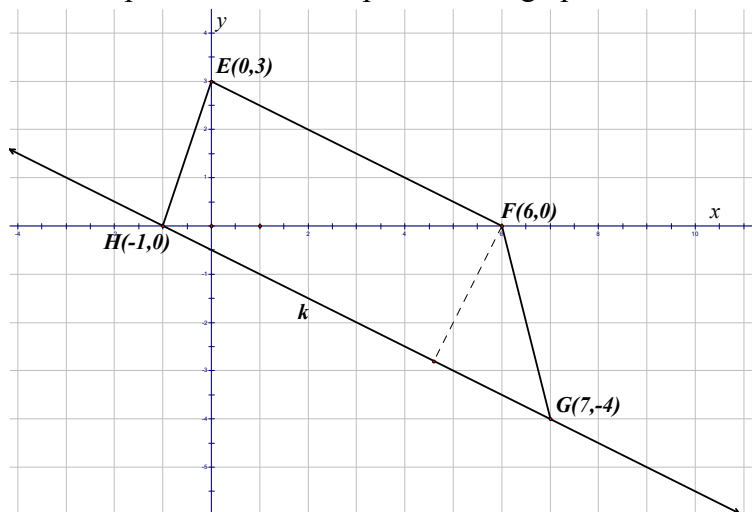
Question 2 (Start a new page)

Marks

The coordinates of the points E , F and G are $(0,3)$, $(6,0)$ and $(7,-4)$ respectively.

- (a) Draw a neat sketch, clearly showing the above information **and** show that the line k which is parallel to EF and passes through point G , has the equation $x + 2y + 1 = 0$.

3



$$\begin{aligned}\text{slope } EF &= \frac{3-0}{0-6} \\ &= -\frac{1}{2}\end{aligned}$$

$$\text{Eqn. } GH : y + 4 = -\frac{1}{2}(x - 7)$$

$$2y + 8 = -x + 7$$

$$x + 2y + 1 = 0$$

- (b) Find the coordinates of the point H where line k meets the x -axis. Clearly mark point H and the position line k on your diagram.

2

$$\text{when } y = 0, \quad x + 1 = 0$$

$$\therefore x = -1$$

$$H \text{ is } (-1,0)$$

- (c) Find the exact length of EF .

2

$$\begin{aligned}EF &= \sqrt{3^2 + 6^2} \\ &= \sqrt{45}\end{aligned}$$

$$\text{length } EF = 3\sqrt{5} \text{ units}$$

- (d) Find perpendicular distance from point F to line GH .

2

$$\begin{aligned}\perp \text{ dist.} &= \frac{|6 + 0 + 1|}{\sqrt{1^2 + 2^2}} \text{ units} \\ &= \frac{7}{\sqrt{5}} \text{ units} \\ &= \frac{7\sqrt{5}}{5} \text{ units}\end{aligned}$$

- (e) Find the exact area of quadrilateral $EFGH$.

$$\begin{aligned}GH &= \sqrt{8^2 + (-4)^2} \\ &= \sqrt{80}\end{aligned}$$

$$\text{length } GH = 4\sqrt{5} \text{ units}$$

$$\begin{aligned}\text{Area} &= \frac{1}{2} \left(\frac{7}{\sqrt{5}} \right) (3\sqrt{5} + 4\sqrt{5}) \text{ u}^2 \\ &= 24.5 \text{ u}^2\end{aligned}$$

Question 3 (Start a new page)

	Marks
(a) Differentiate with respect to x :	
(i) $\frac{4x^3\sqrt{x} + 5}{x^2}$,	2
$f(x) = 4x^{1.5} + 5x^{-2}$	
$f'(x) = 6x^{0.5} - 10x^{-3}$	
$= 6\sqrt{x} - \frac{10}{x^3}$	
(ii) $(8 - 3x)^5$.	2
$f(x) = (8 - 3x)^5$	
$f'(x) = 5(8 - 3x)^4 \times -3$	
$= -15(8 - 3x)^4$	
(b) Find the equation of the tangent to $y = xe^{-x}$ at the point where $x = 2$. Write your answer in general form.	4
$y = xe^{-x}$	
$y' = (1)(e^{-x}) + (x)(-e^{-x})$	
$= (1 - x)e^{-x}$	
When $x = 2, y = 2e^{-2}$	
When $x = 2, y' = -e^{-2}$	
Tangent is: $y - 2e^{-2} = -e^{-2}(x - 2)$	
$y - 2e^{-2} = -e^{-2}x + 2e^{-2}$	
$e^{-2}x + y - 4e^{-2} = 0$	
$x + e^2y - 4 = 0$	
(c) Differentiate $y = \frac{\cos x}{1 + \sin x}$, hence show that $\frac{dy}{dx} = \frac{-1}{1 + \sin x}$	4
$\frac{dy}{dx} = \frac{(1 + \sin x)(-\sin x) - (\cos x)(\cos x)}{(1 + \sin x)^2}$	
$= \frac{-\sin x - \sin^2 x - \cos^2 x}{(1 + \sin x)^2}$	
$= \frac{-\sin x - (\sin^2 x + \cos^2 x)}{(1 + \sin x)^2}$	
$= \frac{-\sin x - 1}{(1 + \sin x)^2}$	
$= \frac{-(\sin x + 1)}{(1 + \sin x)^2}$	
$\therefore \frac{dy}{dx} = \frac{-1}{1 + \sin x}$	

Question 4 (Start a new page)

(a) Evaluate

Marks

$$\begin{aligned}
 \text{(i)} \quad \int_0^4 \frac{dx}{3x+2} &= \frac{1}{3} [\ln(3x+2)]_0^4 \\
 &= \frac{1}{3} (\ln 14 - \ln 2) \\
 &= \frac{\ln 7}{3}
 \end{aligned}$$

2

$$\begin{aligned}
 \text{(ii)} \quad \int_{-1}^2 \cos\left(\frac{\pi}{2}x\right) dx &= \frac{2}{\pi} \left[\sin\left(\frac{\pi}{2}x\right) \right]_{-1}^2 \\
 &= \frac{2}{\pi} \left\{ \sin \pi - \sin\left(-\frac{\pi}{2}\right) \right\} \\
 &= \frac{2}{\pi} \{0 + 1\} \\
 &= \frac{2}{\pi}
 \end{aligned}$$

2

(b) (i) Find the coordinates of the stationary points on the curve $y = (x+1)(x^2-8)$ and determine their nature.

5

$$\begin{aligned}
 y &= (x+1)(x^2-8) \\
 &= x^3 + x^2 - 8x - 8
 \end{aligned}$$

$$\begin{aligned}
 y' &= 3x^2 + 2x - 8 \\
 &= (3x-4)(x+2)
 \end{aligned}$$

for stat. point $y' = 0$

$$\therefore (3x-4)(x+2) = 0$$

$$x = \frac{4}{3} \text{ or } -2$$

$$\begin{aligned}
 \text{if } x = \frac{4}{3}, \quad y &= \left(\frac{4}{3}\right)^3 + \left(\frac{4}{3}\right)^2 - 8\left(\frac{4}{3}\right) - 8 \\
 &= -14\frac{14}{27}
 \end{aligned}$$

$$\begin{aligned}
 \text{if } x = -2, \quad y &= (-2)^3 + (-2)^2 - 8(-2) - 8 \\
 &= 4
 \end{aligned}$$

test nature of stat. points

$$y'' = 6x + 2$$

$$\begin{aligned}
 \text{when } x = -2, \quad y'' &= 6(-2) + 2 \\
 &= -10 < 0
 \end{aligned}$$

$\therefore$ concave down, $\therefore$ local max. t.p. at $(-2, 4)$

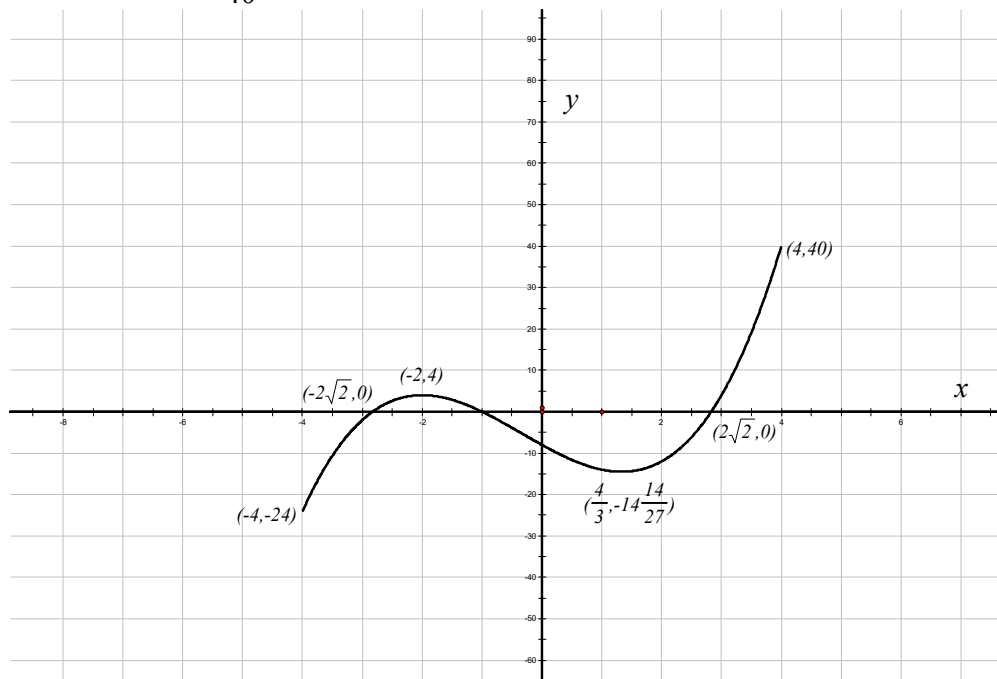
$$\begin{aligned}
 \text{when } x = \frac{4}{3}, \quad y'' &= 6\left(\frac{4}{3}\right) + 2 \\
 &= 10 > 0
 \end{aligned}$$

$\therefore$ concave up, $\therefore$ local min. t.p. at $\left(\frac{4}{3}, -14\frac{14}{27}\right)$

- (ii) Draw a neat **half page** sketch of $y = (x+1)(x^2 - 8)$ in the domain $-4 \leq x \leq 4$.
On your diagram clearly indicate the coordinates and the positions of the intercepts with the coordinate axes and the stationary points.

$$\begin{aligned} \text{when } x = -4, y &= (-4)^3 + (-4)^2 - 8(-4) - 8 \\ &= -24 \end{aligned}$$

$$\begin{aligned} \text{when } x = 4, y &= (4)^3 + (4)^2 - 8(4) - 8 \\ &= 40 \end{aligned}$$



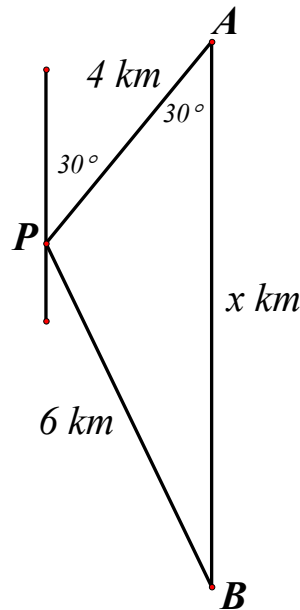
Question 5 (Start a new page)

Marks

- (a) Town A is 4 km from town P and its bearing from town P is $030^\circ T$. Town B is due south of town A and 6 km from town P .

- (i) Draw a neat sketch that clearly illustrates the above information.

2



- (ii) Find the distance between towns A and B . Give your answer correct to the nearest kilometre.

4

Let $AB = x$ km

$$6^2 = 4^2 + x^2 - 2(4)(x)\cos 30^\circ$$

$$36 = 16 + x^2 - 8\left(\frac{\sqrt{3}}{2}\right)x$$

$$36 = 16 + x^2 - 4\sqrt{3}x$$

$$x^2 - 4\sqrt{3}x - 20 = 0$$

$$x = \frac{4\sqrt{3} \pm \sqrt{(4\sqrt{3})^2 - 4(1)(-20)}}{2}$$

$$x = \frac{4\sqrt{3} \pm \sqrt{48 + 80}}{2}$$

$$x = \frac{4\sqrt{3} \pm \sqrt{128}}{2}$$

$$x = \frac{4\sqrt{3} \pm 8\sqrt{2}}{2}$$

$$x = 2\sqrt{3} \pm 4\sqrt{2}$$

$$\text{distance} = (2\sqrt{3} + 4\sqrt{2}) \text{ km since dist.} > 0$$

$$\text{distance} = 9 \text{ km (to nearest km)}$$

OR

Let perpendicular distance from P to AB be y km and let the perpendicular from P meet AB at C .

$$\frac{AC}{4} = \cos 30^\circ$$

$$AC = 4\cos 30^\circ$$

$$= 2\sqrt{3}$$

$$\frac{PC}{4} = \sin 30^\circ$$

$$PC = 4\sin 30^\circ$$

$$= 2$$

$$PC^2 + BC^2 = PB^2$$

$$BC^2 = 6^2 - 2^2$$

$$= 32$$

$$PC = 4\sqrt{2}$$

$$AB = AC + BC$$

$$= 2\sqrt{3} + 4\sqrt{2}$$

$$\therefore \text{distance} = (2\sqrt{3} + 4\sqrt{2}) \text{ km}$$

- (b) (i) Find the intersection points of the line $y = 2x$ and the parabola $y = 6x - x^2$.

2

$$6x - x^2 = 2x$$

$$x^2 - 4x = 0$$

$$x(x - 4) = 0$$

$$x = 0 \text{ or } 4$$

$$x = 0, y = 0 \Rightarrow (0, 0)$$

$$x = 4, y = 8 \Rightarrow (4, 8)$$

- (ii) Find the area bounded by the line $y = 2x$ and the parabola $y = 6x - x^2$.

4

$$A = \int_0^4 [(6x - x^2) - 2x] dx \quad \text{or} \quad A = \int_0^4 (6x - x^2) dx - \frac{1}{2} \times 4 \times 8$$

$$= \int_0^4 (4x - x^2) dx$$

$$= \left[2x^2 - \frac{1}{3}x^3 \right]_0^4$$

$$= \left(2 \times 4^2 - \frac{1}{3} \times 4^3 \right) - 0$$

$$= 10\frac{2}{3}$$

$$\text{Area} = 10\frac{2}{3} \text{ u}^2$$

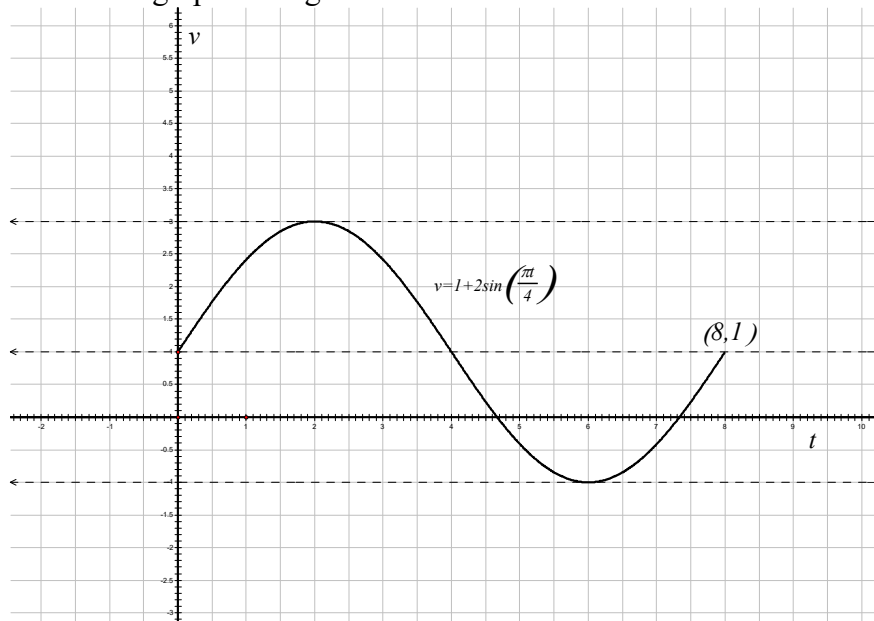
Question 6 (Start a new page)

Marks

- (a) A particle is moving along a straight line. The particle is initially at the origin O and at time t minutes its velocity, v m/min, is given by $v = 1 + 2\sin\left(\frac{\pi}{4}t\right)$.

- (i) Sketch the graph of v against t for $0 \leq t \leq 8$.

2



- (ii) Find the position of the particle when it first reaches its maximum speed.

4

Max. speed when $t = 2$ (from turning point of v - t graph)

$$x = t - \frac{8}{\pi} \cos\left(\frac{\pi t}{4}\right) + c$$

when $t = 0, x = 0$

$$\therefore 0 = 0 - \frac{8}{\pi} \cos(0) + c$$

$$c = \frac{8}{\pi}$$

$$\therefore x = t - \frac{8}{\pi} \cos\left(\frac{\pi t}{4}\right) + \frac{8}{\pi}$$

when $t = 2$

$$x = 2 - \frac{8}{\pi} \cos\left(\frac{2\pi}{4}\right) + \frac{8}{\pi}$$

$$= 2 + \frac{8}{\pi}$$

$$\text{disp.} = \left(2 + \frac{8}{\pi}\right) \text{ cm}$$

- (b) The gradient function of a curve is given by $\frac{dy}{dx} = 3 + \frac{10}{\sqrt{x}}$.

3

Find the equation of the curve if it passes through the point $P(4,9)$.

$$\frac{dy}{dx} = 3 + 10x^{-\frac{1}{2}}$$

$$y = 3x + 20\sqrt{x} + c$$

$$\text{at } P(4,9)$$

$$9 = 3 \times 4 + 20\sqrt{4} + c$$

$$c = -43$$

$$y = 3x + 20\sqrt{x} - 43$$

- (c) (i) Show that the discriminant Δ for $x^2 + mx + m$ is given by $\Delta = m^2 - 4m$.

1

$$\begin{aligned}\Delta &= m^2 - 4(1)(m) \\ &= m^2 - 4m\end{aligned}$$

- (ii) Hence, or otherwise, find the values of k for which the line $y = x + k$ and the curve $y = \frac{x}{x+1}$ do not intersect.

2

Curves intersect when

$$\frac{x}{x+1} = x + k$$

$$x = (x+1)(x+k)$$

$$x = x^2 + kx + x + k$$

$$x^2 + kx + k = 0$$

for no points of intersection the above equation must have no real roots
i.e. $\Delta < 0$

$$\text{Now } \Delta = k^2 - 4k$$

$$\therefore k^2 - 4k < 0$$

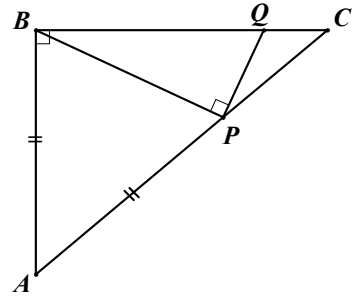
$$k(k-4) < 0$$

$$\therefore 0 < k < 4$$

Question 7 (Start a new page)

Marks

- (a) $\triangle ABC$ and $\triangle BPQ$ are right-angled triangles and $AB = AP$.



- (i) Copy the diagram onto your answer sheet and prove that $\angle CBP = \angle CPQ$.
(Hint: Let $\angle CBP = \alpha^\circ$)

3

Let $\angle CBP = \alpha^\circ$

$\angle ABP + \alpha^\circ = 90^\circ$ (angle sum of right angle $ABC = 90^\circ$)

$\angle ABP = 90^\circ - \alpha^\circ$

$\angle APB = 90^\circ - \alpha^\circ$ (equal angles are opposite equal sides in $\triangle ABP$)

$\angle CPQ + 90^\circ + (90^\circ - \alpha^\circ) = 180^\circ$ (angle sum of straight angle $APC = 180^\circ$)

$\angle CPQ = \alpha^\circ$

$\therefore \angle CBP = \angle CPQ$ (both $= \alpha^\circ$)

- (ii) Hence prove that $PC^2 = BC \times QC$

3

In $\triangle BPC$ and $\triangle PQC$

$\angle CBP = \angle CPQ$ (from (i))

$\angle BCP = \angle PCQ$ (common)

$\therefore \triangle BPC \sim \triangle PQC$ (equiangular)

$\frac{PC}{QC} = \frac{BC}{PC}$ (ratios of corresponding sides of similar triangles are equal)

$\therefore PC^2 = BC \times QC$

* Also possible to prove that a circle passes through points B , P and Q and that APC is also a tangent to the circle, then use theorems about tangents and secants (3 unit theorem)

- (b) A mass of wire cable is wound around a winch. The mass, M kg, of wire on the winch, at time t minutes, is given by the formula $M = 400 - 25\sqrt{t + 100}$.

- (i) Find the initial mass of wire on the winch.

2

when $t = 0$, $M = 400 - 25\sqrt{100}$
 $= 150$

Initial mass = 150Kg

- (ii) Find the time needed to remove all the wire from the winch.

2

when $M = 0$,

$$0 = 400 - 25\sqrt{t+100}$$

$$25\sqrt{t+100} = 400$$

$$\sqrt{t+100} = 16$$

$$t+100 = 256$$

$$t = 156$$

$$\text{time} = 156 \text{ minutes } (= 2\text{hrs } 36\text{min})$$

- (iii) Find the initial rate at which the wire is being removed from the winch.

2

$$\frac{dM}{dt} = -25 \times \frac{1}{2}(t+100)^{-\frac{1}{2}}$$

$$= \frac{-25}{2\sqrt{t+100}}$$

when $t = 0$

$$\frac{dM}{dt} = \frac{-25}{2\sqrt{100}}$$

$$= -1.25$$

$$\text{rate} = 1.25 \text{ kg/min}$$

Question 8 (Start a new page)

Marks

- (a) John starts work on an annual salary of \$38 500. On the last day of each year for the first 10 years of employment, he will receive an annual salary increase of \$1 200. For his remaining years of employment he will receive an annual increase of \$1 650.

- (i) What will John's annual salary be during his 6th year of employment?

2

Years 1 to 10

$\{38\,500, 38\,500 + 1 \times 1\,200, 38\,500 + 2 \times 1\,200, 38\,500 + 3 \times 1\,200, \dots\}$

$$A_6 = 38\,500 + 5 \times 1\,200$$

$$= 44\,500$$

$$\therefore \text{Salary} = \$44\,500$$

- (ii) What will John's annual salary be during his 15th year of employment?

2

Years 1 to 10

$\{38\,500, 38\,500 + 1 \times 1\,200, 38\,500 + 2 \times 1\,200, \dots, 38\,500 + 9 \times 1\,200 (= 49\,300)\}$

Years 11 to 25

$\{49\,300 + 1 \times 1\,650, 49\,300 + 2 \times 1\,650, 49\,300 + 3 \times 1\,650, 49\,300 + 4 \times 1\,650, \dots\}$

$$A_{10} = 38\,500 + 9 \times 1\,200$$

$$= 49\,300$$

$$A_{15} = 49\,300 + 5 \times 1\,650$$

$$= 57\,550$$

$$\therefore \text{Salary} = \$57\,550$$

- (iii) How much will John's total earnings amount to at the end of his 25th year of employment?

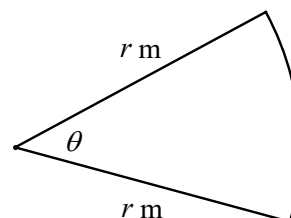
2

$$T = \frac{10}{2} \{2(38\,500) + 9(1\,200)\} + \frac{15}{2} \{2(49\,300) + 14(1\,650)\}$$

$$= 1\,351\,750$$

$$\text{Total earnings} = \$1\,351\,750$$

- (b) A wire framework enclosing an area of 144 m^2 is to be made in the shape of a sector of a circle (see diagram). The length of wire required is L metres where the circle has radius r metres and the sector angle is θ radians.



- (i) Show that $L = 2r + \frac{288}{r}$.

2

$$L = 2r + r\theta$$

$$\text{but } \frac{1}{2}r^2\theta = 144$$

$$\therefore \theta = \frac{288}{r^2}$$

$$L = 2r + r \times \frac{288}{r^2}$$

$$L = 2r + \frac{288}{r}$$

- (ii) Find the minimum length of wire needed to make the framework.

4

$$L = 2r + \frac{288}{r}$$

$$= 2r + 288r^{-1}$$

$$\frac{dL}{dr} = 2 - 288r^{-2}$$

$$= 2 - \frac{288}{r^2}$$

$$\text{for stat. point } \frac{dL}{dr} = 0$$

$$\frac{288}{r^2} = 2$$

$$2r^2 = 288$$

$$r^2 = 144$$

$$r = 12 \text{ (radius } > 0)$$

test nature of stat. point

$$\frac{d^2L}{dr^2} = \frac{576}{r^3}$$

when $r = 12$

$$\frac{d^2L}{dr^2} = \frac{576}{12^3} > 0$$

$\therefore$ concave up, $\therefore$ local min. t.p.

Since the curve is continuous for $r > 0$ and there is only one turning point then the local minimum is the absolute minimum

$$\text{when } r = 12, L = 2(12) + \frac{288}{12}$$

$$= 48$$

$\therefore$ Minimum length = 48m

Question 9 (Start a new page)

	Marks
(a) Point T is 6 cm from the centre O of a circle with radius 3 cm. Tangents drawn from point T touch the circle at points M and N and the tangents are perpendicular to the radii at these points of contact.	
(i) Find the exact size of $\angle MOT$.	2
(ii) Find the exact area of the minor sector MON .	3
(b) (i) Sketch the curve $y = 2 + \sqrt{x}$ and clearly shade the area bounded by the curve, the coordinate axes and the line $x = 9$.	2
(ii) Find the exact volume of the solid formed when the above area is rotated one revolution about the y -axis.	5

Question 10 (Start a new page)

	Marks
(a) In order to enter a university each student is required to sit for a theory examination. If a student passes the theory examination they are then required to sit for a practical examination. It is estimated that the probability for a student to pass the theory examination is 0.8 and to pass the practical examination is 0.75.	
(i) Find the probability that a student chosen at random will gain entry to the university.	2
(ii) Find the probability that if two students chosen at random only one will gain entry to the university.	2
(b) At 6am a population was observed to be undergoing exponential growth and after time t hours the number of individuals N_G was given by $N_G = 100e^{0.3t}$. When the population reached 500, a virus attacked the population and from that time the population underwent exponential decay so that t hours after the entry of the virus, the number of individuals N_D present was given by $N_D = 500e^{kt}$ for some constant $k < 0$. When the population fell to its original value it was decreasing at a rate of 15 individuals per hour.	
(i) At what time of the day did the population reach 500? Give your answer correct to the nearest minute.	3
(ii) Find the value of the constant k .	2
(iii) At what time of the day did the population return to its original value?	3

This is the END of the examination paper